BHUPESH RAJ JOSHI (E.I.T)

Civil/Transportation Engineer

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OBJECTIVE To excel in the field of civil engineering and always support the advanced methodologies and technology to help sustainable development and management of resources. Currently having an E.I.T, I am motivated to get a P.E in future.

EDUCATION The University of Toledo, Toledo, OH **August 2017**
MS in Civil Engineering (Transportation & Traffic Engineering) GPA 3.95
Thesis: "Development of a Methodology for Prioritization of Transportation Projects involving Multiple Assets."

Pulchowk Campus, Tribhuvan University, Nepal **Jan 2014**
BS in Civil Engineering GPA 3.6

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, The University of Toledo, Toledo, OH **August 2015 – Current**

- Worked on development of web based decision support tools for transportation asset management.
  - Project Level trade-off tool using AHP to rank projects using data driven results, update & review optimization tools in network and project level using visual basic.NET, SQL server, MS-access & excel.
- City of Toledo project: Pavement evaluation and preservation project by collection of pavement condition data, analyzing and evaluating PCR using PCR Rating manual from ODOT.

Civil Engineer/Designer, DRUFCON Consultants, Kathmandu, Nepal **Jan 2014 – Jul 2015**

- Surveying and preparing engineering drawings & topographical maps using AutoCAD Civil 3D
- Site supervision, Estimation & costing, Tender/Bid documents, Technical report writing, Bid evaluation.
- Roadway alignment, profile & X-section using Civil 3D and SW-DTM, involved in contractor-client meetings, mentoring interns, volunteering for Rapid Visual Damage Assessment of earthquake affected buildings in Nepal.

RELEVANT ACADEMIC PROJECTS

- Design of an isolated 4-way intersection using SIDRA intersection software **Fall 2016**
  - Collection of existing and previous traffic data, traffic signal design and capacity analysis using a lane based model with performance measures like delays, queue length, demand/capacity ratio from HCM2000.
- Analysis of causes of traffic collisions and severity using ArcGIS and R-statistical tool. **Spring 2016**
  - Used results from logistic regression using R to show collision hotspots and severity in GIS map.

TECHNICAL SKILLS AND SOFTWARE

- Estimation & costing, GIS mapping & spatial analysis, watershed management, soil tests and investigation, landfill design, surveying, land development, construction management.
- Roadway geometric design, traffic signal design, impact studies, capacity analysis, crash studies, transportation asset management, intelligent transportation systems (ITS), data collection, operations and research.

Software and tools:

- Civil/Transportation design: AutoCAD Civil 3D/Micro-station, ArcGIS, Sketch-up, Total-Station, HEC-RAS
- Traffic design, simulation & modeling: SIDRA, HCS, SYNCHRO/SimTraffic (Basic), VISSIM (Basic)
- Programming: SQL server, .NET (visual basic), MS Excel/Access, MATLAB, R-studio for statistical analysis.
- Engineering manuals: HCM, MUTCD, AASHTO Green Book

AWARDS

- Graduate Research Assistantship at University of Toledo during Master's degree.
- Merit based scholarship from Government of Nepal during Bachelor's degree in Civil Engineering.

RELEVANT CERTIFICATIONS/MEMBERSHIPS

- Engineer in Training (EIT/FE) Michigan board, American Society of Civil Engineers (ASCE), Institute of Transportation Engineers (ITE), Nepal Engineers' Association (NEA), Nepal Engineering Council (NEC)

RELEVANT COURSES

- GIS Applications, Transportation and Traffic Engineering, Advanced Engineering Systems Modeling, Soil Mechanics and Foundation, Surveying, Structural analysis, RCC design, Steel & Timber Structures, Hydraulics, Watershed management, Environmental Impact Assessment (EIA), Linear Statistical Model.